

Levi Davis

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PROFESSIONAL EXPERIENCE

Neuroimaging Data Analyst

Nov 2023 – Present

- Design and maintain end-to-end neuroimaging analysis pipelines in Python and Bash on an HPC cluster, processing 10,000+ multimodal MRIs (T1w, T2, FLAIR, DTI, QSM, SWI) to rigorous QA and reproducibility standards.
- Build machine learning and deep learning models (PyTorch, TensorFlow, scikit-learn) for brain and lesion segmentation, disease classification, and predictive modeling; co-authored a published radiomic analysis linking thalamic structure to cognition in Multiple Sclerosis.
- Engineer multimodal MRI registration and preprocessing workflows (ANTs, FreeSurfer, FSL, SynthStrip), handling anisotropic acquisitions and cross-modality alignment across large-scale datasets.
- Manage data infrastructure for 10,000+ scans, including automated transfer, de-identification, and BIDS-compliant organization; led migration of imaging pipelines to new on-prem and AWS HPC clusters.
- Operate a low-field MRI scanner (64mT Hyperfine Swoop), overseeing acquisition protocols and participant QA for active clinical research studies.
- Serve as technical point of contact across research and clinical teams (Epic, REDCap); mentor undergraduate and high school students in neuroimaging methods and analysis workflows.

Data Science Intern

May 2023 – Aug 2023

- Developed time-series forecasting models (ARIMA, LSTM) in R to project solar energy generation, accurately capturing seasonal trends and shifts in statewide supply and demand.

Research Assistant

Mar 2022 – Jul 2022

- Produced 3D scientific animations (Houdini FX) modeling the neuroanatomical progression of Alzheimer's disease for the Animating Alzheimer's documentary project.
- Self-taught the animation software from scratch to deliver publication- and film-ready visualizations of disease progression.

Technical Research Intern

Oct 2021 – May 2022

- Collaborated across multiple institutions to prepare data visualizations for the Virginia Solar Survey, using R and Excel for data wrangling and geographic/temporal analysis.

EDUCATION

University of Virginia — Master of Data Science

May 2023

University of Virginia — B.A. Cognitive Science

May 2022

Concentrations in Neuroscience and Computer Science; Minor in Applied Mathematics

Continuing Education: BCI Spring School (2022–2025); Biomedical Image and Signal Processing and Learning (2024)

PUBLICATIONS

A Radiomic Approach to Clinical MRI Refines the Thalamus-Cognition Link in Multiple Sclerosis. Journal of Neuroimaging, 2025. (Co-author) — doi.org/10.1111/jon.70069

Population Age Group Sensitivity for COVID-19 Infections with Deep Learning. arXiv, 2023. (Co-author, Master's Capstone) — doi.org/10.48550/arXiv.2307.00751

PROJECTS

RadiomicViz (github.com/levijb/radiomicviz, 2026) — Open-source Python package for ROI-level and voxelwise radiomic feature extraction from multimodal MRI; preset-driven configuration, parallelized batch processing, SLURM cluster tooling, and an interactive 3D radiomic feature viewer.

LEADERSHIP EXPERIENCE

Career Services Chair, University of Virginia School of Data Science — Oct 2022 – May 2023

Adaptive Ski Instructor, Therapeutic Adventures Adaptive Skiing, Charlottesville, VA — Winter 2019, 2020

Math Teacher, Flagway™ Program, Young People's Project, Alton, IL — Sep 2017 – May 2018

High School Tutor (Volunteer), Alton, IL — 2016 – 2019

TECHNICAL SKILLS

Languages & Tools: Python, R, MATLAB, Bash, SQL, Linux/Unix, Git, AWS

ML/AI: PyTorch, TensorFlow, Scikit-learn, Deep Learning, Statistical ML, Bayesian ML

Imaging: MRI (T1w, T2, FLAIR, DTI, QSM, SWI), Radiomics, Registration, Segmentation

Neuroimaging Tools: FSL, FreeSurfer, SPM, ANTs, 3D Slicer, ITK-SNAP, NiBabel

Data & Methods: Mixed-Effects Models, Time-Series Analysis (ARIMA/LSTM), Clustering, Statistical Testing, Cross-Validation, BIDS, Epic/REDCap